

Unit-I Tutorial

1. no. of kg. from A = x_1
 no. of kg. from B = x_2

Objective function: $\text{Min } z = 5x_1 + 6x_2$

Constraints:

$$\begin{aligned} x_1 + x_2 &\geq 5 \\ x_1 &\geq 2 \\ x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$$

LPP: $\text{Min } z = 5x_1 + 6x_2$
 s.t $\begin{aligned} x_1 + x_2 &\geq 5 \\ x_1 &\geq 2 \\ x_2 &\leq 4 \\ x_1, x_2 &\geq 0 \end{aligned}$

	M_1	M_2	Availability
Grinding	4	2	$2 \times 40 = 80$
Polishing	2	5	$3 \times 60 = 180$
Profit	₹3	₹4	

Let x_1 & x_2 be the no. of units in M_1 & M_2 .

Objective function: $\text{Max } z = 3x_1 + 6x_2$

Constraints:

Grinder is $4x_1 + 2x_2 \leq 80$

Polisher is $2x_1 + 5x_2 \leq 180$

$$x_1, x_2 \geq 0$$

LPP: $\text{Max } z = 3x_1 + 6x_2$
 s.t $4x_1 + 2x_2 \leq 80$
 $2x_1 + 5x_2 \leq 180$
 $x_1, x_2 \geq 0$

3. Given $\text{Max } z = 300x_1 + 400x_2$

① $5x_1 + 4x_2 = 200$

$$\Rightarrow \frac{x_1}{40} + \frac{x_2}{50} = 1$$

$$x_1 \rightarrow 40$$

$$x_2 \rightarrow 50$$

② $3x_1 + 5x_2 = 150$

$$\Rightarrow \frac{x_1}{50} + \frac{x_2}{30} = 1$$

$$x_1 \rightarrow 50$$

$$x_2 \rightarrow 30$$

③ $5x_1 + 4x_2 = 100$

$$\Rightarrow \frac{x_1}{20} + \frac{x_2}{25} = 1$$

$$x_1 \rightarrow 20$$

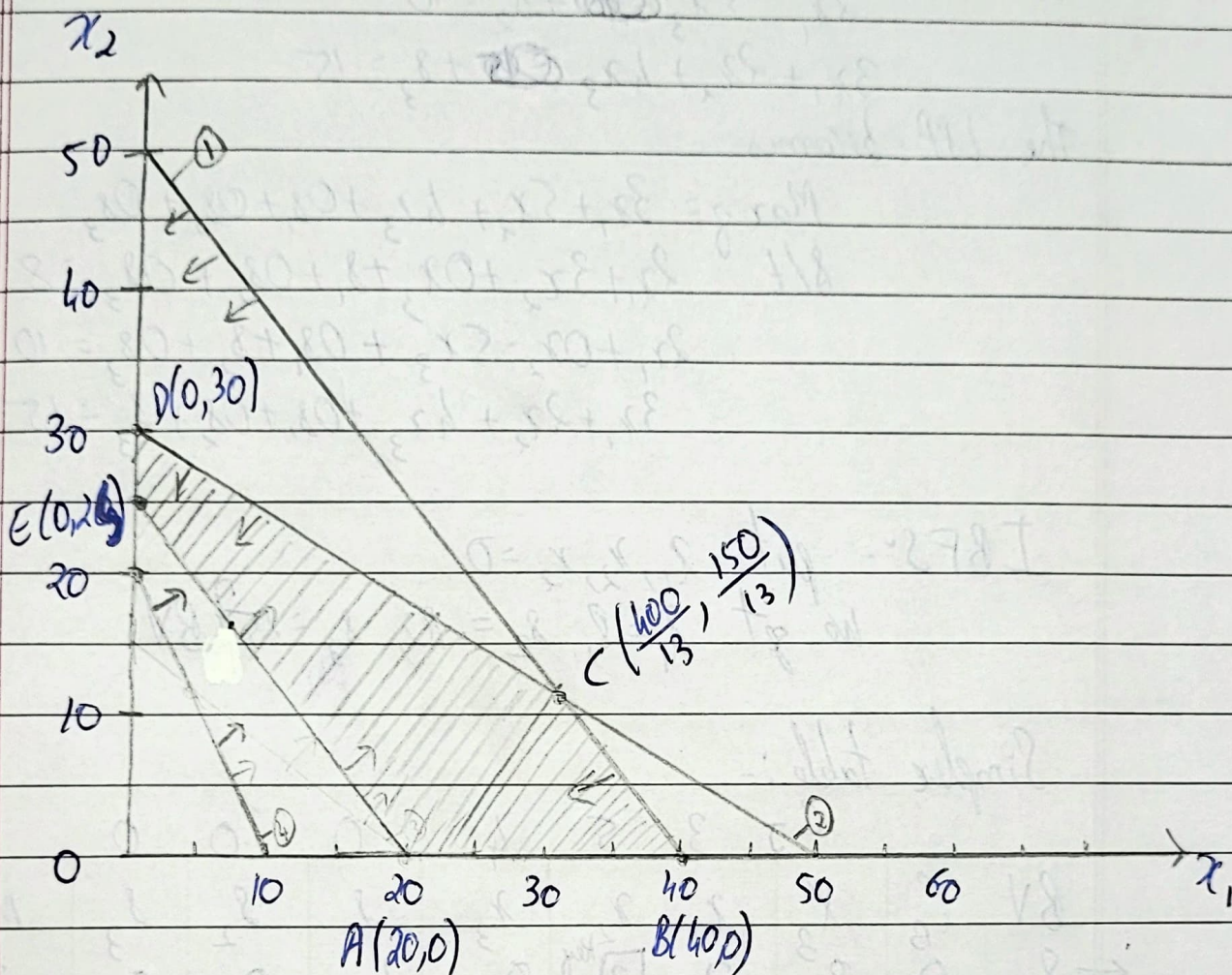
$$x_2 \rightarrow 25$$

$$\textcircled{1} \quad 8x_1 + 4x_2 = 80$$

$$\Rightarrow \frac{x_1}{10} + \frac{x_2}{20} = 1$$

$$x_1 \rightarrow 10$$

$$x_2 \rightarrow 20$$



$$Z(A) = 300(20) + 400(0) = 6000$$

$$Z(B) = 300(40) + 400(0) = 12000$$

$$Z(C) = 300\left(\frac{400}{13}\right) + 400\left(\frac{150}{13}\right) = 13846.1538$$

$$Z(D) = 300(0) + 400(30) = 12000$$

$$Z(E) = 300(0) + 400(20) = 8000$$

$$Z_{\max} \text{ at } C\left(\frac{400}{13}, \frac{150}{13}\right)$$

$$\text{Optimal solution: } x_1 = \frac{400}{13}, x_2 = \frac{150}{13}, Z_{\max} = \frac{180000}{13}$$

4. Given Max $z = 3x_1 + 5x_2 + 4x_3$

Introducing slack variable,

$$2x_1 + 3x_2 + s_1 = 8$$

$$2x_1 - 5x_3 + s_2 = 10$$

$$3x_1 + 2x_2 + 4x_3 + s_3 = 15$$

The LPP becomes

$$\text{Max } z = 3x_1 + 5x_2 + 4x_3 + 0s_1 + 0s_2 + 0s_3$$

$$\text{s.t. } 2x_1 + 3x_2 + 0x_3 + s_1 + 0s_2 + 0s_3 = 8$$

$$2x_1 + 0x_2 - 5x_3 + 0s_1 + s_2 + 0s_3 = 10$$

$$3x_1 + 2x_2 + 4x_3 + 0s_1 + 0s_2 + s_3 = 15$$

IBFS:- put $x_1, x_2, x_3 = 0$

we get $s_1 = 8, s_2 = 10, s_3 = 15$ BV

Simplex table:-

	C_j		3	5	4	0	0	0	
BV	C_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3	M/R
$\leftarrow s_1$	0	8	2	3 ^{key}	0	1	0	0	$2/3 \rightarrow$
s_2	0	10	2	0	-5	0	1	0	X
s_3	0	15	3	2	4	0	0	1	7.5
			-3	-5 [↑]	-4	0	0	0	
x_2	5	$9/3$	$2/3$	1	0	$1/3$	0	0	X
s_2	0	10	2	0	-5	0	1	0	X
$\leftarrow s_3$	0	$29/3$	$5/3$	0	4 ^{key}	$-2/3$	0	1	$29/12 \rightarrow$
			$1/3$	0	-4 [↑]	$5/3$	0	0	

x_2	5	$\frac{8}{3}$	$\frac{2}{3}$	1	0	$\frac{1}{3}$	0	0
s_2	0	$\frac{265}{12}$	$\frac{49}{12}$	0	0	$-\frac{5}{6}$	1	$\frac{5}{4}$
x_3	4	$\frac{29}{12}$	$\frac{5}{12}$	0	1	$-\frac{1}{6}$	0	$\frac{1}{4}$
			2	0	0	1	0	1

Here all $\Delta_i \geq 0$, hence the solution is optimal
 Optimal solution: $x_1 = 0, x_2 = \frac{8}{3}, x_3 = \frac{29}{12}$
 $Z_{\max} = 23$

5. Given Max $z = 2x_1 + 4x_2$
 Introducing slack, surplus & AV,

$$2x_1 + 3x_2 + s_1 = 48$$

$$x_1 + 3x_2^{+s_2} = 42$$

$$x_1 + x_2 - s_3 + A = 21$$

The given LPP becomes

$$\text{Max } z = 2x_1 + 4x_2 + 0s_1 + 0s_2 + 0s_3 - MA$$

$$\text{s.t. } 2x_1 + 3x_2 + s_1 + 0s_2 + 0s_3 + 0A = 48$$

$$x_1 + 3x_2 + 0s_1 + s_2 + 0s_3 + 0A = 42$$

$$x_1 + x_2 + 0s_1 + 0s_2 - s_3 + A = 21$$

IBFS:- put $x_1, x_2, s_3 = 0$

we get $s_1 = 48$

$s_2 = 42$

$A = 21$

} BV

Simplex table:-

BV	C_B	X_B	2	4	0	0	0	-M	M/R
s_1	0	48	2	3	1	0	0	0	16
$\leftarrow s_2$	0	42	1	$\boxed{3}^{key}$	0	1	0	0	14 $\rightarrow$
A	-M	21	1	1	0	0	-1	1	21
			-M-2	$\boxed{-M-4}^{\uparrow}$	0	0	M	0	
$\leftarrow s_1$	0	6	$\boxed{1}^{key}$	0	1	-1	0	0	6 $\rightarrow$
x_2	4	14	$\frac{1}{3}$	1	0	$\frac{1}{3}$	0	0	42
A	-M	7	$\frac{2}{3}$	0	0	$-\frac{1}{3}$	-1	1	10.5
			$\boxed{\frac{-2M-2}{3}}^{\uparrow}$	0	0	$\frac{M+4}{3}$	M	0	
x_1	2	6	1	0	1	-1	0	0	x
x_2	4	12	0	1	$-\frac{1}{3}$	$\frac{2}{3}$	0	0	18
$\leftarrow A$	-M	3	0	0	$-\frac{2}{3}$	$\boxed{\frac{1}{3}}^{key}$	-1	1	9 $\rightarrow$
			0	0	$\frac{2M+2}{3}$	$\boxed{\frac{-M+2}{3}}^{\uparrow}$	M	0	
x_1	2	15	1	0	-1	0	-3	3	
x_2	4	6	0	1	1	0	2	$-\frac{2}{3}$	
s_2	0	9	0	0	-2	1	-3	3	
			0	0	2	0	2	$\frac{M+10}{3}$	

All $\Delta_i \geq 0$, solution is optimal.
 Optimal solution: $x_1 = 15, x_2 = 6$
 $Z_{max} = 54$

6. Given $\text{Max } z = 2x_1 + x_2 + x_3$
 Introducing slack, surplus & AV,
 $4x_1 + 6x_2 + 3x_3 + s_1 = 8$
 $3x_1 - 6x_2 - 4x_3 + s_2 = 1$
 $2x_1 + 3x_2 - 5x_3 - 8A = 4$

Phase - I :

$\text{Max } z^* = 0x_1 + 0x_2 + 0x_3 + 0s_1 + 0s_2 + 0s_3 - A$
 $s.t. \quad 4x_1 + 6x_2 + 3x_3 + s_1 + 0s_2 + 0s_3 + 0A = 8$
 $3x_1 - 6x_2 - 4x_3 + 0s_1 + s_2 + 0s_3 + 0A = 1$
 $2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 - 8s_3 + A = 4$

IBFS: put $x_1, x_2, x_3, s_3 = 0$

we get $s_1 = 8, s_2 = 1, A = 4$ } BV

Simplex table:-

			0	0	0	0	0	0	-1	
BV	C_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3	A	M/R
s_1	0	8	4	6	3	1	0	0	0	$4/3$
s_2	0	1	3	-6	-4	0	1	0	0	x
$\leftarrow A$	-1	4	2	$\boxed{3}^{K_0}$	-5	0	0	-1	1	$4/3 \rightarrow$
			-2	$\boxed{-3}^{K_1}$	5	0	0	1	0	
s_1	0	0	0	0	13	1	0	2	-2	
s_2	0	9	7	0	-14	0	1	-2	2	
x_2	0	$1/3$	$2/3$	1	$-5/3$	0	0	$-1/3$	$1/3$	
			0	0	0	0	0	0	1	

Here Δ 's ≥ 0 .
We proceed to phase II.

Phase II:-

			2	1	1	0	0	0	
BV	C_B	X_B	x_1	x_2	x_3	s_1	s_2	s_3	m/R
$\leftarrow s_1$	0	0	0	0	$\boxed{13}$ key	1	0	2	0 $\rightarrow$
s_2	0	9	7	0	-14	0	1	-2	x
x_2	1	$4/3$	$2/3$	1	$-5/3$	0	0	$-1/3$	x
			$-4/3$	0	$\boxed{-8/3}$ $\uparrow$	0	0	$-1/3$	
x_3	1	0	0	0	1	$1/3$	0	$2/3$	x
$\leftarrow s_2$	0	9	$\boxed{7}$ key	0	0	$14/3$	1	$2/3$	1.2857 $\rightarrow$
x_2	1	$4/3$	$2/3$	1	0	$5/39$	0	$-1/13$	2
			$\boxed{-4/3}$ $\uparrow$	0	0	$5/39$	0	$-1/13$	
x_3	1	0	0	0	1	$1/3$	0	$2/3$	
x_4	2	$9/2$	1	0	0	$2/13$	$1/7$	$\frac{2}{9}$	
x_2	1	$10/21$	0	1	0	$1/39$	$-2/21$	$-25/273$	
			0	0	0	$16/39$	$4/21$	$29/273$	

Here Δ 's ≥ 0 hence the solution is optimal.
Optimal solution: $x_1 = 9/2, x_2 = 10/21, x_3 = 0$
 $\sum_{max} = 64/21$